

PIIM — Prediction Input Identification Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Basis Standard (PBAS)

Operational Layer: Prediction Basis Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Input Identification

Version: 1.0

Status: Canonical · Open Module

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction Input Identification

Canonical Definition

Prediction Input Identification Module (PIIM) defines the governance conditions under which information, data, observations, signals, measurements, records, outputs, judgments, and other informational elements that materially enter or inform a Prediction Methodology are identified as Prediction Inputs.

PIIM establishes the first governed informational layer of the Prediction Basis.

It distinguishes:



Information that exists

from:

information that is available

from:

information that is actually used or materially informs the prediction.

PIIM does not determine whether an identified input is sufficiently relevant, trustworthy, independent, evidentially valid, or sufficient to support the prediction.

Its purpose is more fundamental:

the informational elements entering the prediction must first become identifiable.

Operational Role

PIIM serves as the first internal governance space of the Prediction Basis Standard (PBAS) .

The module asks:

What information actually enters or materially informs the prediction, and in what input form does it participate?

The governed progression begins:

Available / Accessible Information → Material Input Identification → Input Classification → Input State → Identified Prediction Inputs

The resulting identified Prediction Inputs become an input to:

PSMM — Predictive Source Mapping Module.

Internal Module Flow

→ Prediction Input Identification → Predictive Source Mapping → Signal Relevance → Input Sufficiency → Prediction Basis Traceability

Current Position: Prediction Input Identification

Module Operational Space

PIIM governs:

Prediction Input identity; input recognition; input inclusion; direct inputs; indirect inputs; derived inputs; transformed inputs; aggregated inputs; human-provided inputs; machine-generated inputs; AI-generated inputs; model-generated inputs; simulation-derived inputs; market-derived inputs; observed inputs; measured inputs; historical inputs; current inputs; real-time inputs; contextual inputs; external inputs; internal inputs; static inputs; dynamic inputs; material input states; input availability; input use; unavailable expected inputs; excluded potential inputs; undocumented inputs; hidden inputs; input identity changes; and input identification traceability.

Module Function

PIIM prevents predictive systems from operating with an informational foundation that cannot be reconstructed.

A prediction may be generated from hundreds, thousands, or millions of informational elements.

Yet governance fails at the first stage if an organization cannot establish which categories or materially relevant elements actually entered or informed the prediction.

PIIM therefore creates the transition:

Information Environment → Identified Prediction Inputs

It prevents:

“the system had access to this information”

from automatically becoming:

“the prediction was based on this information.”

Information Access $\neq$ Prediction Input

Prediction Input

A Prediction Input is an identifiable informational element, informational state, or governed input category that materially enters or informs the Prediction Methodology used to construct a prediction.

Prediction Inputs may include:

numerical data; text; images; sensor measurements; historical records; current observations; expert judgments; human assessments; market information; economic indicators; operational signals; model outputs; AI-generated outputs; simulation outputs; probabilities; classifications; scores; event records; environmental conditions; contextual information; other materially relevant informational elements.

The existence of information does not make it a Prediction Input.

Information Environment

A predictive system may operate within a much larger Information Environment than the Prediction Basis it actually uses.

The Information Environment may contain:

Available Information + Accessible Information + Observed Information + Ignored Information + Excluded Information + Used Information

PIIM identifies which elements materially cross the boundary into Prediction Input status.

Information Environment $\neq$ Prediction Basis

Input Recognition

An informational element becomes identifiable as a Prediction Input where it materially participates in the predictive process.

Participation may occur through:

direct ingestion; retrieval; observation; transformation; human interpretation; model processing; rule application; aggregation; contextual conditioning; manual adjustment; another materially relevant mechanism.

PIIM requires such participation to become visible where material.

Direct Prediction Inputs

Direct Prediction Inputs enter Prediction Methodology without an intermediate predictive or analytical output acting as their immediate input form.

Examples may include:

sensor readings; economic measurements; event records; prices; weather observations; operational measurements.

Directness describes the input relationship.

It does not establish source originality or evidentiary quality.

Direct Input $\neq$ Original Source by Default

Indirect Prediction Inputs

An indirect input reaches the predictive process through another informational or analytical layer.

Examples may include:

research summaries; analyst reports; external scores; derived indicators; third-party model outputs; aggregated market measures.

PIIM identifies the input as indirect where this distinction is materially relevant.

PSMM subsequently governs its source relationships.

Derived Prediction Inputs

Derived inputs are produced from one or more preceding informational elements.

Examples include:

calculated ratios; composite indicators; extracted features; generated scores; summarized variables; inferred states.

PIIM identifies their participation as Prediction Inputs.

Source lineage and dependency are subsequently governed through PBAS modules where applicable.

Derived Input $\neq$ Independent Information by Default

Transformed Prediction Inputs

Information may undergo transformation before entering the operative predictive process.

Transformations may include:

normalization; scaling; encoding; filtering; cleaning; categorization; feature extraction; aggregation; summarization.

PIIM preserves the fact that the transformed representation, rather than necessarily the original raw information, enters the prediction.

PMES retains ownership of methodological transformation logic.

Aggregated Prediction Inputs

Multiple informational elements may enter Prediction Methodology through a single aggregated input.

Examples may include:

indexes; composite scores; sentiment measures; market indicators; aggregated sensor states.

The aggregated form must remain distinguishable from its constituent information where that distinction is material.

One Aggregated Input $\neq$ One Underlying Information Source

Observed and Measured Inputs

Prediction Inputs may originate from direct observation or measurement.

PIIM identifies the input form without independently determining whether the observation or measurement accurately represents Operational Reality.

Observed Input $\neq$ Governed Operational Reality

ORA™ retains governance of Operational Reality.

Human-Provided Inputs

Humans may provide:

expert judgments; classifications; estimates; observations; qualitative assessments; scores; adjustments; contextual information.

Where these materially inform Prediction Methodology, they become identifiable Prediction Inputs.

Human Judgment $\neq$ Established Fact

The governance of cognition, responsibility, evidence, or accountability remains outside PIIM.

Machine-Generated Inputs

Machine-generated information may become Prediction Input where it materially enters the predictive process.

Examples include:

algorithmic classifications; automated scores; model outputs; extracted features; generated indicators; machine-detected events.

PIIM preserves machine-generated origin as an input characteristic where material.

AI-Generated Inputs

AI-generated information must remain distinguishable from directly observed information.

An AI system may generate:

summaries; inferred variables; classifications; synthetic representations; predictions used by another predictor; contextual assessments.

Where such outputs materially enter Prediction Methodology, PIIM identifies them as Prediction Inputs.

AI-Generated Input $\neq$ Observed Reality

Model-Generated Inputs

One model's output may become another predictive system's input.

This creates a layered predictive relationship:

Underlying Information $\rightarrow$ Model A $\rightarrow$ Model Output $\rightarrow$ Prediction Input to Methodology B

PIIM identifies the model output as a Prediction Input.

It does not assume that the output constitutes an independent source.

Simulation-Derived Inputs

Simulation outputs may become Prediction Inputs.

Examples include:

simulated trajectories; scenario frequencies; simulated failure states; modeled system responses; synthetic operational conditions.

PIIM requires their simulation-derived nature to remain identifiable.

Simulation-Derived Input $\neq$ Operational Observation

SIMULOS® retains governance ownership of the simulation itself.

Market-Derived Inputs

Prediction Inputs may include:

market prices; prediction-market prices; trading volume; market-implied probabilities; order-book information; volatility; market sentiment.

PIIM identifies these as market-derived inputs where applicable.

It does not determine their predictive relevance or whether a market-implied probability constitutes a governed probability estimate.

Market Price $\neq$ Governed Probability Estimate

Historical Prediction Inputs

Historical information may enter Prediction Methodology as input.

PIIM identifies:

the historical period; the relevant informational form; the relationship to the prediction formation point where material.

Historical information is not automatically relevant merely because it is available.

SRVM governs relevance.

Current and Real-Time Inputs

Predictive methodologies may consume current or real-time information.

PIIM distinguishes these input forms where timing materially affects their identity.

A real-time input should not be confused with a historical observation simply because both contain the same variable.

Contextual Inputs

Some information does not directly represent the Prediction Target but materially conditions how other information is interpreted.

Examples may include:

regulatory conditions; environmental conditions; operating regime; geographic context; market state; system configuration.

Where such information materially informs Prediction Methodology, it may constitute a contextual Prediction Input.

Static and Dynamic Inputs

Static inputs remain substantially fixed during the relevant prediction process.

Dynamic inputs may change continuously or periodically.

The distinction becomes material where input state or timing affects reconstruction of Prediction Basis.

Same Input Variable $\neq$ Same Input State

Input Availability

PIIM distinguishes whether a potential input was:

Available

Unavailable

Partially Available

Delayed

Unknown

at the relevant prediction point.

Availability does not establish actual use.

Available Information $\neq$ Used Prediction Input

Input Use

An input is treated as used where it materially enters or informs Prediction Methodology.

PIIM therefore distinguishes:

Available

from:

Accessed

from:

Used

where the distinction is material.

Accessed Information $\neq$ Used Prediction Input

Excluded Potential Inputs

Information may be available but deliberately excluded from the predictive process.

PIIM identifies material exclusion where necessary to distinguish the Prediction Basis from the broader Information Environment.

It does not determine whether exclusion was methodologically justified.

That may become relevant to subsequent PBAS governance.

Unavailable Expected Inputs

A methodology may expect an input that is unavailable at prediction time.

PIIM preserves this state where material.

An unavailable expected input must not silently disappear from the informational description of the predictive process.

Missing Input $\neq$ Nonexistent Input Requirement

Undocumented Inputs

An input may materially influence a prediction without being formally documented.

Examples may include:

manual expert adjustments; informal external information; hidden system variables; undocumented contextual judgments.

PIIM treats such conditions as an identification problem requiring clarification.

Undocumented Input $\neq$ Non-Input

Hidden Inputs

Predictive systems may contain informational dependencies that are not visible to the immediate user.

Examples may include:

upstream service outputs; hidden contextual variables;

automatically retrieved information; embedded model-generated states; external system signals.

Where these materially influence prediction formation, their existence should become governable.

Input Granularity

Not every individual data point must necessarily become a separately governed object.

PIIM permits appropriate levels of input identification such as:

individual element; variable; dataset; information class; input stream; governed input category.

The appropriate granularity depends upon what is necessary to understand and reconstruct the Prediction Basis.

Governance Granularity $\neq$ Maximum Possible Granularity

Input Materiality

PIIM focuses on informational elements capable of materially affecting the predictive process.

Incidental technical information need not automatically become a governed Prediction Input.

Materiality may depend upon whether the information:

enters predictive logic; conditions another input; affects model behavior; influences human judgment; changes predictive output; changes interpretation of the predictive process.

Input State

An identified Prediction Input may possess a governed state.

Relevant states may include:

Expected

Available

Unavailable

Accessed

Used

Excluded

Replaced

Superseded

The state describes the input's relationship to the prediction process.

Input Identity Change

An input may change materially while retaining the same label.

For example:

a dataset may be updated; a sensor may be replaced; an external index may change methodology; a model output may originate from a new model version; an expert assessment may be revised.

PIIM requires materially changed input identity or state to remain distinguishable where necessary.

Same Input Name $\neq$ Same Input State

Prediction Input Timing

The timing of input availability and use may materially affect Prediction Basis.

PIIM should preserve whether an input was available:

before prediction formation,

during prediction formation,

or:

only afterward

where relevant.

Post-Prediction Information $\neq$ Original Prediction Input

This prevents hindsight from contaminating historical Prediction Basis reconstruction.

Input Identification Sufficiency

Prediction Input Identification is sufficiently governed when:

material inputs are identifiable; relevant input categories are distinguishable; direct, indirect, derived, or transformed status is identifiable where material; human-, machine-, AI-, simulation-, and market-derived inputs remain distinguishable where relevant; input availability and actual use can be separated; material exclusions and unavailable expected inputs remain visible; undocumented material inputs trigger clarification; material input timing can be reconstructed; historical input states are not silently overwritten.

PIIM does not require relevance or sufficiency determination.

Those belong to later PBAS modules.

Governance Requirements

PIIM requires that:

1. material Prediction Inputs be identifiable; 2. available information be distinguishable from information actually used;

3. accessed information not automatically be treated as used input; 4. direct and indirect inputs remain distinguishable where material; 5. derived and transformed inputs remain identifiable; 6. aggregated inputs not obscure materially relevant underlying input structure; 7. human-provided inputs remain identifiable where they materially affect prediction; 8. machine- and AI-generated inputs remain distinguishable from observed information; 9. simulation-derived inputs remain distinguishable from Operational Reality observations; 10. market-derived inputs remain identifiable as such; 11. material contextual inputs remain visible; 12. unavailable expected inputs remain identifiable; 13. material exclusions remain distinguishable from unavailable information; 14. undocumented or hidden material inputs trigger clarification; 15. input timing remain identifiable where material; 16. materially changed input states remain distinguishable; 17. historical input identity not be overwritten by later information states.

Governance Logic

PIIM applies the following governance logic:

Predictive Information Environment

↓

Potential Informational Elements Identified

↓

Material Participation Assessed

↓

Prediction Inputs Recognized

↓

Input Form Identified

↓

Direct / Indirect / Derived / Transformed / Aggregated Status Identified where applicable

↓

Human / Machine / AI / Simulation / Market / Observational Origin Type Identified where applicable

↓

Availability / Access / Use / Exclusion State Identified

↓

Input Timing Established

↓

Material Input Identity Preserved

↓

Identified Prediction Inputs

↓

PSMM — Predictive Source Mapping Module

Where material input identity cannot be established:

Prediction Input Identification Gap → Clarification / Reconstruction / Disclosure Requirement

Minimum Implementation Framework

A minimum implementation of PIIM must establish:

A. Prediction Reference

The prediction or governed predictive process to which the inputs relate.

B. Input Identifier

A sufficiently unique identifier for each material input or governed input category.

C. Input Description

What the informational element represents.

D. Input Form

Where applicable:

Direct / Indirect / Derived / Transformed / Aggregated / Contextual

E. Input Origin Type

Where applicable:

Observed / Measured / Human / Machine / AI / Model / Simulation / Market / External / Other

F. Availability State

At minimum:

Available / Unavailable / Partial / Delayed / Unknown

G. Participation State

At minimum:

Expected → Available → Accessed → Used

with applicable alternative states:

Excluded / Unavailable / Replaced / Superseded

H. Input Timing

The time or relevant temporal state at which the input became available and materially participated in the prediction.

I. Materiality

Whether the input materially participates in Prediction Methodology.

J. Transformation Status

Whether the input represents original, transformed, derived, or aggregated information.

K. Identification Gap

Any material input whose identity, form, participation, or timing cannot be sufficiently established.

L. Historical Preservation

The ability to preserve the input identity and state associated with the prediction at formation time.

Governance Outputs

PIIM may produce:

Prediction Input Register; identified Prediction Inputs; input classification; input form record; input origin-type record; input availability state; input participation state; excluded-input record; unavailable-input record; undocumented-input flag; hidden-input clarification requirement; input timing record; input materiality record; historical input-state record; unresolved input-identification requirements.

The primary governed output is:

Identified Prediction Inputs

This becomes the principal input into:

PSMM — Predictive Source Mapping Module.

Critical Distinctions

Information Environment ≠ Prediction Basis

Information Access ≠ Prediction Input

Available Information ≠ Governed Prediction Input

Available Information ≠ Used Prediction Input

Accessed Information ≠ Used Prediction Input

Direct Input ≠ Original Source by Default

Derived Input ≠ Independent Information by Default

One Aggregated Input ≠ One Underlying Information Source

Observed Input ≠ Governed Operational Reality

Human Judgment ≠ Established Fact

AI-Generated Input ≠ Observed Reality
Simulation-Derived Input ≠ Operational Observation
Market Price ≠ Governed Probability Estimate
Same Input Variable ≠ Same Input State
Missing Input ≠ Nonexistent Input Requirement
Undocumented Input ≠ Non-Input
Governance Granularity ≠ Maximum Possible Granularity
Same Input Name ≠ Same Input State
Post-Prediction Information ≠ Original Prediction Input

Enterprise Application

PIIM provides organizations with a controlled inventory of the information that actually enters predictive systems.

This becomes increasingly important where predictions are produced through complex combinations of: enterprise data + external APIs + AI systems + human expertise + models + simulations + market information + real-time operational signals.

Without Prediction Input Identification, an organization may know which predictive system produced an output while remaining unable to determine what information materially informed that output.

PIIM creates the first informational accountability point:

Before asking whether an input was good, relevant, sufficient, or trustworthy, establish whether it was actually an input.

Use Case 1 — AI Prediction Uses Information Nobody Knew It Was Accessing

A company deploys an AI system to predict customer demand.

The official methodology documentation states that the prediction uses:

historical sales; inventory; seasonal patterns; pricing; marketing activity.

Later investigation reveals that the AI workflow also automatically retrieves external web information and incorporates third-party market summaries through an upstream service.

These inputs were never listed in the Prediction Basis.

The forecasts appear accurate, but the organization cannot reconstruct which external information materially influenced particular predictions.

The critical governance question is:

Can an organization claim to know the basis of an AI prediction when material information entered the predictive process without being identified as Prediction Inputs?

PIIM requires those informational dependencies to become visible before their sources, relevance, or sufficiency are assessed.

Undocumented Input ≠ Non-Input

Use Case 2 — A Prediction Is Judged Using Information That Did Not Exist Yet

An investment team makes a prediction that a company will fail to complete a major acquisition.

Three weeks later, new regulatory information becomes public and strongly supports the original prediction.

During retrospective review, analysts cite that later information as evidence that the original forecast had been well founded.

However, the information was not available when the prediction was formed.

The critical governance question is:

Can information that became available only after a prediction was made legitimately be represented as part of the informational basis of that earlier prediction?

PIIM requires input timing to remain identifiable.

The later information may become relevant to monitoring or revision.

It cannot become an original Prediction Input retrospectively.

Post-Prediction Information $\neq$ Original Prediction Input

Canonical Boundary

PIIM governs the identification, classification, participation state, timing, materiality, and historical preservation of informational elements that materially enter or inform Prediction Methodology. It does not govern detailed source relationships, source dependency, Predictor–Reality Proximity, signal relevance, Prediction Basis sufficiency, evidence validity, predictive assumptions, Prediction Formation, probability representation, Operational Reality, or predictive performance.

PIIM establishes:

what information actually entered or materially informed the prediction.

It does not determine:

whether that information was sufficient to justify the prediction.

Canonical Closing Statement

A prediction cannot have a governed informational foundation while the information that actually entered it remains unknown.

PIIM makes the inputs visible before PGA™ determines where they came from, whether they matter, and whether they are sufficient.

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